

Xu-Cheng He

ACADEMY RESEARCH FELLOW / SUB-GROUP LEADER

University of Helsinki

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Education

University of Helsinki

PHD ATMOSPHERIC SCIENCE (DISTINCTION)

• Advisors: Prof. Markku Kulmala, Prof. Jasper Kirkby

Finland

2018.01 - 2021.09

University of Helsinki

MSc ATMOSPHERIC SCIENCE

Finland

2015.08 - 2017.10

Yunnan University

BSc ATMOSPHERIC SCIENCE

China

2011.09 - 2015.07

Professional Experience

2025.01-Present **Sub-group leader and co-PI for the CLOUD project**, University of Helsinki
2023.01-2024.12 **Visiting Researcher (simulation of aerosols)**, University of Cambridge
2017.09-2024.12 **CLOUD project coordinator for marine runs (3 mos. per year)**, University of Helsinki and CERN
2022.03-2022.12 **Visiting Researcher (global model training)**, Carnegie Mellon University and Finnish Meteorological Institute
2020.09-2021.08 **Visiting Researcher (operation of urban observatory)**, Nanjing University and Beijing University of Chemical Technology
2016.06-2017.12 **Research Assistant**, University of Helsinki

Awards

2023 **Invited participant of the Atmospheric Chemistry Colloquium for Emerging Senior Scientists (ACCESS XVII)**, Brookhaven National Laboratory
2022 **Extraordinary Potential Prize of 2021 Chinese Government Award for Outstanding Self-financed Students Abroad (20 awardees globally)**, Ministry of Education, China \$10,000
2022 **Dissertation prize, all faculties (4/500)**, University of Helsinki €4,000
2022 **NOSA ECS Aerosologist award (Early Career Scientist)**, Nordic Society for Aerosol Research
2021 **Outstanding thesis award**, Doctoral school in Natural Sciences, University of Helsinki €2,000
2015 **International student grant**, University of Helsinki €1,500
2015 **Best thesis award (1/43)**, Yunnan University
2012 **Campus star in science and innovation (10/4000)**, Yunnan University ¥2,000

Fellowships & Grants

2025-2029 **Academy research fellowship**, Research Council of Finland / University of Helsinki €1,126,320
2025-2026 **co-PI, Constraining the size distribution and chlorine production of ferric chloride aerosols for quantitative atmospheric methane removal**, Spark Climate Solutions \$299,248
2022-2025 **Postdoctoral researcher fellowship**, Research Council of Finland / University of Helsinki €364,377
2022 **Postdoctoral research grant**, Jenny and Antti Wihuri foundation €56,000
2018-2021 **Doctoral school fellowship**, University of Helsinki ca. €108,000

Key publications

PUBLICATIONS AS A **CO-FIRST[%]** OR **CORRESPONDING AUTHOR[#]**

Summary: 5 manuscripts as (co-)first-author and 5 as last-author, in total 10 as corresponding author, including 2 in *Science* and 1 in *Nature*.

2025

R. Zhang, H.-B. Xie[#], F. Ma, R. Yin, J. Chen, **X.-C. He**, Quaternary Nucleation of Iodine and Sulfur Oxoacids in the Marine Atmosphere: Unexpected Role of Methanesulfonic Acid. **Journal of Geophysical Research: Atmosphere**. (2025).

2024

J. Shen[%], D.M. Russell[%], J. DeVivo^{,...}, J. Kirkby[#], J. Curtius[#], **X.-C. He[#]**, New particle formation from isoprene under upper tropospheric conditions. **Nature**. (2024).

B. Rörup, **X.-C. He[#]**, J. Shen^{,...}, R. Volkamer, D. Worsnop, K. Lehtipalo, Temperature, humidity, and ionisation effect of iodine oxoacid nucleation. **Environmental Science: Atmosphere**. (2024).

Y. Zhang[%], D. Li[%], **X.-C. He[#]**, ..., J. Jiang, A. Ding, M. Kulmala, Iodine oxoacids and their roles in sub-3 nanometer particle growth in polluted urban environments. **Atmospheric Chemistry & Physics**. (2024).

2023

X.-C. He[#], M. Simon, S. Iyer, H.-B. Xie[#], ..., N.M. Donahue, M. Sipilä[#], M. Kulmala[#], Iodine oxoacids enhance nucleation of sulfuric acid particles in the atmosphere. **Science**. (2023).

X.-C. He[#], J. Shen[#], S. Iyer^{,...}, J. Mikkilä, M. Sipilä, J. Kangasluoma, Characterisation of gaseous iodine species detection using the multi-scheme chemical ionisation inlet 2 with bromide and nitrate chemical ionisation methods. **Atmospheric Measurement Techniques**. (2023).

F. Ma, H.-B. Xie[#], R. Zhang^{,...}, M. Engsvang, J. Elm, **X.-C. He[#]**, Enhancement of Atmospheric Nucleation Precursors on Iodic Acid Induced Nucleation: Predictive Model and Mechanism. **Environmental Science and Technology**. (2023).

2022

H. Finkenzeller^{%#}, S. Iyer[%], **X.-C. He**, ..., T. Kurten[#], M. Rissanen, R.V. Volkamer[#], The gas-phase formation mechanism of iodic acid as an atmospheric aerosol source. **Nature Chemistry**. (2022).

R. Zhang, H.-B. Xie[#], F. Ma^{,...}, M. Sipilä, M. Kulmala, **X.-C. He[#]**, Critical Role of Iodous Acid in Neutral Iodine Oxoacid Nucleation. **Environmental Science & Technology**. 56, 14166-14177 (2022).

2021

M. Wang[%], **X.-C. He^{%#}**, H. Finkenzeller, S. Iyer, D. Chen^{,...}, M. Rissanen, R. Volkamer, Y. J. Tham[#], N. M. Donahue, M. Sipilä, Measurement of iodine species and sulfuric acid using bromide chemical ionization mass spectrometers. **Atmospheric Measurement Techniques**. 14, 4187-4202 (2021).

X.-C. He[#], Y. J. Tham, L. Dada, M. Wang, H. Finkenzeller^{,...}, N. M. Donahue, R. Volkamer, J. Kirkby[#], D. R. Worsnop, M. Sipilä[#], Role of iodine oxoacids in atmospheric aerosol nucleation. **Science**. 371, 589-595 (2021).

X.-C. He[#], S. Iyer, M. Sipilä, A. Ylisirniö, M. Peltola^{,...}, V.-M. Kerminen, R. C. Flagan, J. Kirkby[#], T. Kurtén, M. Kulmala, Determination of the collision rate coefficient between charged iodic acid clusters and iodic acid using the appearance time method. **Aerosol Science & Technology**. 55, 231-242 (2021).

Y. J. Tham, **X.-C. He**, Q. Li, C. A. Cuevas, J. Shen^{,...}, M. Kulmala, C. O'Dowd, M. Dal Maso, A. Saiz-Lopez[#], M. Sipilä[#], Direct field evidence of autocatalytic iodine release from atmospheric aerosol. **Proceedings of the National Academy of Sciences**. 118 (2021).

2020

M. Wang[%], W. Kong[%], R. Marten, **X.-C. He**, ..., J.H. Seinfeld, I. El-Haddad, R.C. Flagan, N.M. Donahue[#], Rapid growth of new atmospheric particles by nitric acid and ammonia condensation. **Nature**. 581 (2020).

2019

D. Zhao, R. Yang[#], Y. Tao, W.K. Zhang and **X.-C. He**, Objective detection of the Kunming quasi-stationary front. **Theoretical and Applied Climatology**. 138 (2019).

2017

S. Iyer[#], **X.-C. He**, N. Hyyttinen, T. Kurtén[#] and M.P. Rissanen, Computational and Experimental Investigation of the Detection of HO₂ Radical and the Products of Its Reaction with Cyclohexene Ozonolysis Derived RO₂ Radicals by an Iodide-Based Chemical Ionization Mass Spectrometer. **The Journal of Physical Chemistry A**. 121 (2017).

F. Bianchi[#], O. Garmash, **X.-C. He**,..., M. Kulmala, M. Ehn and H. Junninen, The role of highly oxygenated molecules (HOMs) in determining the composition of ambient ions in the boreal forest. **Atmospheric Chemistry and Physics**. 17 (2017).

Total publication count: 52. Total citations: Researchgate (1,988), Google scholar (2,113).

Presentations

SELECTED INVITED TALKS

2025.05. Invited seminar, **University of Reading**, UK

2024.06. Invited seminar, **University of Helsinki**, Finland

2024.05. Invited seminar, **Max Planck Institute for Chemistry**, Germany

2024.04. Invited seminar, **Massachusetts Institute of Technology**, USA

2024.04. Invited seminar, **SOLAS open seminar series**

2022.08. Invited speaker, **Gordon Research Conference**, Italy

2022.05. Invited talk, **Carnegie Mellon University**, USA

2021.01. Invited talk, **Nanjing University**, China

CONTRIBUTED PRESENTATIONS

American Meteorological Society annual meeting (session co-chair, 2025, 2024, 2023)

Atmospheric Chemistry Colloquium for Emerging Senior Scientists (Invited, 2023)

Gordon Research Conference - Molecular and Ionic Clusters (Invited talk, 2022)

Gordon Research Conference - Atmospheric Chemistry (Poster, 2025, 2023)

European Geosciences Union General Assembly (Talk, 2021, 2020)

International Aerosol Conference (Talk, 2023; Poster, 2017)

European Aerosol Conference (Talk, 2024, 2022; Poster, 2019)

International Conference on Nucleation and Atmospheric Aerosols (Talk, 2023; Poster, 2017)

International Conference on Aerosol Cycle (Talk, 2017)

Free Radical Symposium (Poster, 2017)

Surface Ocean Lower Atmosphere Study (Poster, 2019)

Cryosphere and Atmospheric Chemistry (Poster, 2017)

Teaching Experience

2024	Tropospheric Atmospheric Chemistry , Guest Lecturer	University of Helsinki
2024	Synthesis of physical chemistry, experiments, observations and models to understand atmospheric particle formation and climate impact , Guest Lecturer	University of Helsinki
2019	Climate science at high latitudes: eScience for linking Arctic measurements and modeling , Teaching Assistant	University of Helsinki
2018	Formation and growth of atmospheric aerosols , Teaching Assistant	University of Helsinki

Outreach & Professional Development

CONFERENCE CHAIR

2022-present **Formation and impacts of atmospheric aerosols and cloud condensation nuclei: experiment, observation, and modeling**, Co-chair *American Meteorological Society Annual Meeting*

MANUSCRIPT PEER REVIEW

Science, One Earth, Environmental Science & Technology, Environmental Science & Technology Letters, Environmental Science & Technology Air, Atmospheric Chemistry and Physics, Geophysical Research Letters, Journal of Geophysical Research: Atmospheres.

FUNDING REVIEW

National Science Foundation (USA)

SERVICE

2017-present: Coordinator for the marine and polar aerosol formation program at the CLOUD project, CERN

2023-present: Coordinator for the modelling group at the CLOUD project, CERN

Supervision & Mentoring

2018.05-2023.04	Jiali Shen , Co-supervised doctoral student (graduated with a distinction)	University of Helsinki
2023.02-present	Wenjuan Yu , Co-supervised doctoral student	University of Helsinki
2017.09-2024.06	Rima Baalbaki , Mentor for part of her doctoral study since 2021.09	University of Helsinki
2018.05-2024.06	Birte Rörup , Mentor for her doctoral study	University of Helsinki Beijing
2018.09-2022.05	Ying Zhang , Mentor for her M.Sc. degree since 2021.01	University of Chemical Technology Nanjing
2017.09-2024.08	Duzitian Li , Mentor for his B.Sc. and M.Sc. degrees since 2020.10	University

Media Coverage

- Dec 2023 **Chemistry World**, Iodine compounds accelerate cloud formation over oceans and the poles
- Oct 2021 **Sciencepost**, L'émission d'iode par l'océan, une influence inattendue sur la banquise arctique
- June 2021 **Lab Manager**, The Impact of Clouds on Climate Change
- Feb 2021 **The Atlantic**, The Arctic Has a Cloud Problem
- Feb 2021 **SCIENMAG**, Climate research: rapid formation of iodine particles over the Arctic
- Feb 2021 **PHYS.ORG**, CLOUD at CERN reveals the role of iodine acids in atmospheric aerosol formation
- Feb 2021 **ScienceDaily**, How iodine-containing molecules contribute to the formation of atmospheric aerosols, affect climate

Full publications

- [1] Federico Bianchi et al. "The role of highly oxygenated molecules (HOMs) in determining the composition of ambient ions in the boreal forest". In: *Atmospheric Chemistry and Physics* 17.22 (Nov. 20, 2017), pp. 13819–13831. ISSN: 1680-7324. DOI: 10.5194/acp-17-13819-2017. URL: <https://acp.copernicus.org/articles/17/13819/2017/> (visited on 08/18/2021).
- [2] Xu-Cheng He. "From the measurement of halogenated species to iodine particle formation". PhD thesis. Helsinki: University of Helsinki, Aug. 27, 2017. 71 pp. URL: <https://helda.helsinki.fi/handle/10138/229173>.
- [3] Siddharth Iyer et al. "Computational and Experimental Investigation of the Detection of HO₂ Radical and the Products of Its Reaction with Cyclohexene Ozonolysis Derived RO₂ Radicals by an Iodide-Based Chemical Ionization Mass Spectrometer". In: *The Journal of Physical Chemistry A* 121.36 (Sept. 14, 2017), pp. 6778–6789. ISSN: 1089-5639, 1520-5215. DOI: 10.1021/acs.jpca.7b01588. URL: <http://pubs.acs.org/doi/10.1021/acs.jpca.7b01588> (visited on 05/15/2018).
- [4] Katrianne Lehtipalo et al. "Multicomponent new particle formation from sulfuric acid, ammonia, and biogenic vapors". In: *Science Advances* 4.12 (Dec. 2018), eaau5363. ISSN: 2375-2548. DOI: 10.1126/sciadv.aau5363. URL: <http://advances.sciencemag.org/lookup/doi/10.1126/sciadv.aau5363> (visited on 11/24/2019).
- [5] Dominik Stolzenburg et al. "Rapid growth of organic aerosol nanoparticles over a wide tropospheric temperature range". In: *Proceedings of the National Academy of Sciences* 115.37 (Sept. 11, 2018), pp. 9122–9127. ISSN: 0027-8424, 1091-6490. DOI: 10.1073/pnas.1807604115. URL: <http://www.pnas.org/lookup/doi/10.1073/pnas.1807604115> (visited on 12/09/2019).
- [6] Qing Ye et al. "Molecular Composition and Volatility of Nucleated Particles from α -Pinene Oxidation between –50 °C and +25 °C". In: *Environmental Science & Technology* 53.21 (Nov. 5, 2019), pp. 12357–12365. ISSN: 0013-936X, 1520-5851. DOI: 10.1021/acs.est.9b03265. URL: <https://pubs.acs.org/doi/10.1021/acs.est.9b03265> (visited on 09/08/2021).
- [7] Di Zhao et al. "Objective detection of the Kunming quasi-stationary front". In: *Theoretical and Applied Climatology* 138.3 (Nov. 2019), pp. 1405–1418. ISSN: 0177-798X, 1434-4483. DOI: 10.1007/s00704-019-02894-w. URL: <http://link.springer.com/10.1007/s00704-019-02894-w> (visited on 09/08/2021).
- [8] Martin Heinritzi et al. "Molecular understanding of the suppression of new-particle formation by isoprene". In: *Atmospheric Chemistry and Physics* 20.20 (Oct. 20, 2020), pp. 11809–11821. ISSN: 1680-7324. DOI: 10.5194/acp-20-11809-2020. URL: <https://acp.copernicus.org/articles/20/11809/2020/> (visited on 09/08/2021).
- [9] Mario Simon et al. "Molecular understanding of new-particle formation from α -pinene between –50 and +25 °C". In: *Atmospheric Chemistry and Physics* 20.15 (Aug. 3, 2020), pp. 9183–9207. ISSN: 1680-7324. DOI: 10.5194/acp-20-9183-2020. URL: <https://acp.copernicus.org/articles/20/9183/2020/> (visited on 09/08/2021).
- [10] Dominik Stolzenburg et al. "Enhanced growth rate of atmospheric particles from sulfuric acid". In: *Atmospheric Chemistry and Physics* 20.12 (June 25, 2020), pp. 7359–7372. ISSN: 1680-7324. DOI: 10.5194/acp-20-7359-2020. URL: <https://www.atmos-chem-phys.net/20/7359/2020/> (visited on 07/19/2020).

- [11] Mingyi Wang et al. "Photo-oxidation of Aromatic Hydrocarbons Produces Low-Volatility Organic Compounds". In: *Environmental Science & Technology* 54.13 (July 7, 2020), pp. 7911–7921. ISSN: 0013-936X, 1520-5851. DOI: 10.1021/acs.est.0c02100. URL: <https://pubs.acs.org/doi/10.1021/acs.est.0c02100> (visited on 09/08/2021).
- [12] Mingyi Wang et al. "Rapid growth of new atmospheric particles by nitric acid and ammonia condensation". In: *Nature* 581.7807 (May 2020), pp. 184–189. ISSN: 0028-0836, 1476-4687. DOI: 10.1038/s41586-020-2270-4. URL: <http://www.nature.com/articles/s41586-020-2270-4> (visited on 07/19/2020).
- [13] Yonghong Wang et al. "Formation of highly oxygenated organic molecules from chlorine-atom-initiated oxidation of alpha-pinene". In: *Atmospheric Chemistry and Physics* 20.8 (Apr. 30, 2020), pp. 5145–5155. ISSN: 1680-7324. DOI: 10.5194/acp-20-5145-2020. URL: <https://acp.copernicus.org/articles/20/5145/2020/> (visited on 09/08/2021).
- [14] Lisa J. Beck et al. "Differing Mechanisms of New Particle Formation at Two Arctic Sites". In: *Geophysical Research Letters* 48.4 (Feb. 28, 2021). ISSN: 0094-8276, 1944-8007. DOI: 10.1029/2020GL091334. URL: <https://onlinelibrary.wiley.com/doi/10.1029/2020GL091334> (visited on 09/17/2021).
- [15] Runlong Cai et al. "Impacts of coagulation on the appearance time method for new particle growth rate evaluation and their corrections". In: *Atmospheric Chemistry and Physics* 21.3 (Feb. 16, 2021), pp. 2287–2304. ISSN: 1680-7324. DOI: 10.5194/acp-21-2287-2021. URL: <https://acp.copernicus.org/articles/21/2287/2021/> (visited on 09/08/2021).
- [16] Lucía Caudillo et al. "Chemical composition of nanoparticles from α -pinene nucleation and the influence of isoprene and relative humidity at low temperature". In: *Atmospheric Chemistry and Physics* 21.22 (Nov. 25, 2021), pp. 17099–17114. ISSN: 1680-7324. DOI: 10.5194/acp-21-17099-2021. URL: <https://acp.copernicus.org/articles/21/17099/2021/> (visited on 11/25/2021).
- [17] Biwu Chu et al. "Particle growth with photochemical age from new particle formation to haze in the winter of Beijing, China". In: *Science of The Total Environment* 753 (Jan. 2021), p. 142207. ISSN: 00489697. DOI: 10.1016/j.scitotenv.2020.142207. URL: <https://linkinghub.elsevier.com/retrieve/pii/S0048969720357363> (visited on 09/08/2021).
- [18] Xu-Cheng He. "Iodine oxoacids in atmospheric aerosol formation : from chamber simulations to field observations". PhD thesis. Helsinki: University of Helsinki, Aug. 23, 2021. 72 pp. URL: <https://helda.helsinki.fi/handle/10138/332625?locale-attribute=en>.
- [19] Xu-Cheng He et al. "Determination of the collision rate coefficient between charged iodic acid clusters and iodic acid using the appearance time method". In: *Aerosol Science and Technology* 55.2 (Feb. 1, 2021), pp. 231–242. ISSN: 0278-6826. DOI: 10.1080/02786826.2020.1839013. URL: <https://doi.org/10.1080/02786826.2020.1839013>.
- [20] Xu-Cheng He et al. "Role of iodine oxoacids in atmospheric aerosol nucleation". In: *Science* 371.6529 (2021), pp. 589–595. ISSN: 0036-8075. DOI: 10.1126/science.abe0298. URL: <https://science.sciencemag.org/content/371/6529/589>.
- [21] Clémence Rose et al. "Investigation of several proxies to estimate sulfuric acid concentration under volcanic plume conditions". In: *Atmospheric Chemistry and Physics* 21.6 (Mar. 24, 2021), pp. 4541–4560. ISSN: 1680-7324. DOI: 10.5194/acp-21-4541-2021. URL: <https://acp.copernicus.org/articles/21/4541/2021/> (visited on 09/08/2021).
- [22] Mihnea Surdu et al. "Molecular characterization of ultrafine particles using extractive electrospray time-of-flight mass spectrometry". In: *Environmental Science: Atmospheres* (2021), 10.1039/D1EA00050K. ISSN: 2634-3606. DOI: 10.1039/D1EA00050K. URL: <http://xlink.rsc.org/?DOI=D1EA00050K> (visited on 09/08/2021).
- [23] Yee Jun Tham et al. "Direct field evidence of autocatalytic iodine release from atmospheric aerosol". In: *Proceedings of the National Academy of Sciences* 118.4 (Jan. 26, 2021), e2009951118. ISSN: 0027-8424, 1091-6490. DOI: 10.1073/pnas.2009951118. URL: <http://www.pnas.org/lookup/doi/10.1073/pnas.2009951118> (visited on 06/09/2021).
- [24] Mingyi Wang et al. "Measurement of iodine species and sulfuric acid using bromide chemical ionization mass spectrometers". In: *Atmospheric Measurement Techniques* 14.6 (June 7, 2021), pp. 4187–4202. ISSN:

- 1867-8548. DOI: 10.5194/amt-14-4187-2021. URL: <https://amt.copernicus.org/articles/14/4187/2021/> (visited on 06/09/2021).
- [25] Mao Xiao et al. "The driving factors of new particle formation and growth in the polluted boundary layer". In: *Atmospheric Chemistry and Physics* 21.18 (Sept. 27, 2021), pp. 14275–14291. ISSN: 1680-7324. DOI: 10.5194/acp-21-14275-2021. URL: <https://acp.copernicus.org/articles/21/14275/2021/> (visited on 12/13/2021).
- [26] Chao Yan et al. "The Synergistic Role of Sulfuric Acid, Bases, and Oxidized Organics Governing New-Particle Formation in Beijing". In: *Geophysical Research Letters* 48.7 (Apr. 16, 2021). ISSN: 0094-8276, 1944-8007. DOI: 10.1029/2020GL091944. URL: <https://onlinelibrary.wiley.com/doi/10.1029/2020GL091944> (visited on 02/20/2023).
- [27] Lisa J. Beck et al. "Diurnal evolution of negative atmospheric ions above the boreal forest: from ground level to the free troposphere". In: *Atmospheric Chemistry and Physics* 22.13 (July 5, 2022), pp. 8547–8577. ISSN: 1680-7324. DOI: 10.5194/acp-22-8547-2022. URL: <https://acp.copernicus.org/articles/22/8547/2022/> (visited on 07/09/2022).
- [28] Qinyi Li et al. "Role of Iodine Recycling on Sea-Salt Aerosols in the Global Marine Boundary Layer". In: *Geophysical Research Letters* 49.6 (Mar. 28, 2022). ISSN: 0094-8276, 1944-8007. DOI: 10.1029/2021GL097567. URL: <https://onlinelibrary.wiley.com/doi/10.1029/2021GL097567> (visited on 04/25/2022).
- [29] Ruby Marten et al. "Survival of newly formed particles in haze conditions". In: *Environmental Science: Atmospheres* (2022), 10.1039/D2EA00007E. ISSN: 2634-3606. DOI: 10.1039/D2EA00007E. URL: <http://xlink.rsc.org/?DOI=D2EA00007E> (visited on 04/25/2022).
- [30] Jiali Shen et al. "High Gas-Phase Methanesulfonic Acid Production in the OH-Initiated Oxidation of Dimethyl Sulfide at Low Temperatures". In: *Environmental Science & Technology* 56.19 (Oct. 4, 2022), pp. 13931–13944. ISSN: 0013-936X, 1520-5851. DOI: 10.1021/acs.est.2c05154. URL: <https://pubs.acs.org/doi/10.1021/acs.est.2c05154> (visited on 01/30/2023).
- [31] Roseline C. Thakur et al. "An evaluation of new particle formation events in Helsinki during a Baltic Sea cyanobacterial summer bloom". In: *Atmospheric Chemistry and Physics* 22.9 (May 17, 2022), pp. 6365–6391. ISSN: 1680-7324. DOI: 10.5194/acp-22-6365-2022. URL: <https://acp.copernicus.org/articles/22/6365/2022/> (visited on 05/20/2022).
- [32] Mingyi Wang et al. "Synergistic HNO₃–H₂SO₄–NH₃ upper tropospheric particle formation". In: *Nature* 605.7910 (May 19, 2022), pp. 483–489. ISSN: 0028-0836, 1476-4687. DOI: 10.1038/s41586-022-04605-4. URL: <https://www.nature.com/articles/s41586-022-04605-4> (visited on 05/20/2022).
- [33] Yonghong Wang et al. "Molecular Composition of Oxygenated Organic Molecules and Their Contributions to Organic Aerosol in Beijing". In: *Environmental Science & Technology* 56.2 (Jan. 18, 2022), pp. 770–778. ISSN: 0013-936X, 1520-5851. DOI: 10.1021/acs.est.1c05191. URL: <https://pubs.acs.org/doi/10.1021/acs.est.1c05191> (visited on 02/28/2022).
- [34] Chao Yan et al. "The effect of COVID-19 restrictions on atmospheric new particle formation in Beijing". In: *Atmospheric Chemistry and Physics* 22.18 (Sept. 19, 2022), pp. 12207–12220. ISSN: 1680-7324. DOI: 10.5194/acp-22-12207-2022. URL: <https://acp.copernicus.org/articles/22/12207/2022/> (visited on 01/17/2023).
- [35] Rongjie Zhang et al. "Critical Role of Iodous Acid in Neutral Iodine Oxoacid Nucleation". In: *Environmental Science & Technology* 56.19 (Oct. 4, 2022), pp. 14166–14177. ISSN: 0013-936X, 1520-5851. DOI: 10.1021/acs.est.2c04328. URL: <https://pubs.acs.org/doi/10.1021/acs.est.2c04328> (visited on 10/07/2022).
- [36] Lucía Caudillo et al. "An intercomparison study of four different techniques for measuring the chemical composition of nanoparticles". In: *Atmospheric Chemistry and Physics* 23.11 (June 15, 2023), pp. 6613–6631. ISSN: 1680-7324. DOI: 10.5194/acp-23-6613-2023. URL: <https://acp.copernicus.org/articles/23/6613/2023/> (visited on 06/20/2023).
- [37] Lubna Dada et al. "Role of sesquiterpenes in biogenic new particle formation". In: *Science Advances* 9.36 (Sept. 8, 2023), eadi5297. ISSN: 2375-2548. DOI: 10.1126/sciadv.adi5297. URL: <https://www.science.org/doi/10.1126/sciadv.adi5297> (visited on 09/17/2023).

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